

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. Explain how Development (Dev) and Operations (Ops) teams need to improve on how to operate and align based on Dev, Ops and Business View.
 - b. How can financial institutions like Dutch banking trade global corporation ING leverage DevOps practices to enhance their operational efficiency and drive innovation in the competitive banking sector?
 - c. Why does the Waterfall software development model have limitations, and why is the Agile approach often considered more beneficial?
 - d. What are the different types of waste in IT processes. Explain with some examples of each. How do they impact overall business outcomes?

2. **Attempt any two of the following:** 12
 - a. How does API management embody the principles of DevOps through the management of the API lifecycle?
 - b. In test automation, how do incomplete requirements lead to faulty software, and how can adopting an automated and agile approach help address this issue?
 - c. What are the different maturity levels of continuous delivery and how do they reflect the evolving capabilities and practices within an organization's software delivery process?
 - d. Describe some valuable advanced analytics and statistical methods that can provide DevOps practitioners with essential information to drive improvements as operational functions become integrated with development.

3. **Attempt any two of the following:** 12
 - a. In creating a DevOps transformation playbook, what is the importance of identifying the target state, specifically in terms of business goals and drivers, and how does this step contribute to the overall success of the transformation?
 - b. Why are the key objectives and benefits of conducting a Value Stream Mapping workshop important in the context of improving processes and efficiency within an organization?
 - c. What steps should be taken to determine the value proposition delivered by the line of business for the various problems addressed across each customer segment?
 - d. How does minimizing cycle time serve as a core theme for DevOps plays aimed at optimizing the delivery pipeline, and what impact does it have on overall efficiency and effectiveness?

- 4. Attempt any two of the following:** **12**
- a. Why is leveraging existing reference sources, roles, and practices important for practical DevOps, particularly in relation to DevOps and Information Security?
 - b. How does the efficiency and collaboration of F1 racing crews compare to the operations of a modern software factory, and what lessons can be drawn from this comparison to make a case for DevOps ROI?
 - c. What new challenges must DevOps practitioners address to ensure the success of Internet of Things (IoT) projects, and how can these challenges impact the overall implementation and performance of IoT solutions?
 - d. Describe the role containers play in driving innovation within a Cloud-Hosted DevOps platform, and how they enhance the effectiveness of DevOps practices.
- 5. Attempt any two of the following:** **12**
- a. What are the core themes for scaling DevOps in an enterprise environment, and how do these themes contribute to the successful implementation and expansion of DevOps practices across large organizations?
 - b. Which team models are crucial for scaling DevOps in an enterprise, and how do these models improve collaboration and the implementation of DevOps practices across the organization?
 - c. Discuss how common anti-patterns in DevOps transformation impact the success of DevOps adoption in the enterprise, and what strategies can be used to identify and address these issues effectively.
 - d. How can starting with pilot projects facilitate DevOps adoption in the enterprise, and what are the key benefits and challenges associated with using pilot projects to drive broader DevOps implementation?
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